

Notes Parent Functions: Cubic & Cube Root

Strategic X-Values: Use **perfect cubes** (0, 1, 8) and their cube roots (0, 1, 2) — these give you integer points!

Cube Root: $f(x) = \sqrt[3]{x}$

x	y
-8	
-1	
0	
1	
8	

x : 1, 8 $\rightarrow$ y : 1, 2

WIDE “S”

Cubic: $f(x) = x^3$

x	y
-2	
-1	
0	
1	
2	

x : 1, 2 $\rightarrow$ y : 1, 8

TALL “S”

Key Features		
Feature	Cube Root: $f(x) = \sqrt[3]{x}$	Cubic: $f(x) = x^3$
Domain		
Range		
Intercept(s)		
Inc / Dec		

THE BIG DISTINCTION: How to Tell Them Apart

	Cube Root $\sqrt[3]{x}$	Cubic x^3
Pattern	$x: 1, 8 \rightarrow y: 1, 2$	$x: 1, 2 \rightarrow y: 1, 8$
Near origin	Steep (changes fast)	Flat (changes slow)
Far from origin	Flat (levels off)	Steep (shoots up/down)

Memory trick: The **8** is always with the cubing.

- Cube root: 8 goes IN, 2 comes OUT — because $\sqrt[3]{8} = 2$
- Cubic: 2 goes IN, 8 comes OUT — because $2^3 = 8$

Finding Intercepts

y-intercept: Set $x = 0$ and solve for y .

x-intercept: Set $y = 0$ and solve for x .

Find the x -intercept and y -intercept of each function.

1. $f(x) = (x - 3)^3 + 8$

2. $f(x) = \sqrt[3]{x + 8} - 2$

x -int: _____

x -int: _____

y -int: _____

y -int: _____

3. $f(x) = -2(x + 1)^3 + 16$

4. $f(x) = \sqrt[3]{x - 1} + 1$

x -int: _____

x -int: _____

y -int: _____

y -int: _____